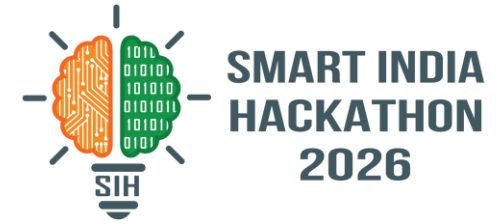


SMART INDIA HACKATHON 2026



RELIEFCHAIN

Problem Statement ID: SW-45

Problem Statement Title: ReliefChain — Blockchain-Verified Disaster Relief Fund Distribution Tracker

Theme: Disaster Management

PS Category: Software

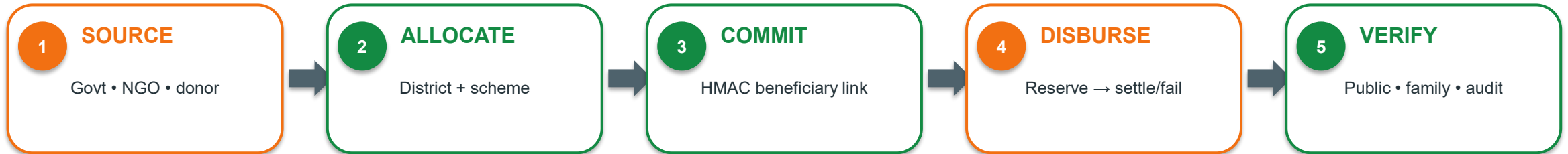
Team ID:

Team Name: PMFC



RELIEFCHAIN

A permissioned evidence trail from fund source to family payout



PROBLEM

- Funds cross many opaque accounts
- Families cannot verify promised aid
- Audits are slow and paper-heavy

SOLUTION

- Fabric records accepted fund transitions
- Role-specific API views preserve privacy
- Committed events drive reconciliation

UNIQUE VALUE

- Trust shifts from a claim to shared evidence
- Public proof without public beneficiary identity
- Ledger–database mismatch becomes detectable

Next.js

Flutter

NestJS

PostgreSQL

Hyperledger Fabric

Docker

TECHNOLOGIES USED • METHODOLOGY & IMPLEMENTATION FLOW

ACTORS

- Public dashboard
- Operator portal
- Beneficiary mobile
- Auditor API

API CONTROL

- JWT + RBAC
- Org ownership
- Eligibility + balances
- Idempotency

PRIVACY

- HMAC identifier
- AES-GCM contacts
- Hashed phone
- Allowlisted events

LEDGER

- 8 Fabric v1 writes
- Stable error codes
- Signed MSP actor
- ReadAsset + history

EVIDENCE

- Committed indexer
- Public proof
- Reconciliation
- CSV / JSON trail

PRIVATE OPERATIONS

PostgreSQL
Encrypted PII • sessions • jobs • attempts

FABRIC CHANNEL
Tamper-evident state • timestamped safe events

FEASIBILITY AND VIABILITY

FEASIBILITY ANALYSIS • POTENTIAL RISKS • MITIGATION STRATEGIES

FEASIBLE NOW

- Working NestJS + PostgreSQL backend
- 3-org Fabric network and deployed chaincode
- Restart-safe demo seed and payout worker
- Health, readiness, audit and proof APIs
- Contract, privacy and invariant test suites

KEY RISKS

- Garbage-in: ledger cannot verify field truth
- Single-host peers are not institutionally independent
- Identity, OTP and payout providers are simulated
- Ledger commit and DB write are not one ACID action
- Key custody and certificate rotation need hardening

MITIGATION PATH

- Regulated UIDAI/payment-provider attestations
- Independent peers + multi-org endorsement
- Outbox/recovery and projection reconciliation
- HSM-backed keys + Fabric CA lifecycle
- Phased district pilot with audit checkpoints

IMPACT AND BENEFITS

TARGET-AUDIENCE IMPACT • SOCIAL, ECONOMIC & OPERATIONAL BENEFITS

BENEFICIARIES

- Private OTP access
- Own payout status
- Opaque proof reference

PUBLIC

- District/scheme totals
- No public recipient PII
- Proof lookup by reference

AUDITORS

- Searchable event trail
- Faster reconciliation
- Export + direct ledger check

GOVT + NGOs

- Attributable actions
- Balance/ownership rules
- Shared source of evidence

Expected Beneficial Outcomes

1

TRACEABLE

Source → allocation → payout

2

TAMPER-EVIDENT

History cannot be silently rewritten

3

PRIVACY-SAFE

Evidence without identity exposure

4

REUSABLE

New disaster, district or scheme

1

Hyperledger Fabric — security model, identities and permissioned governancehyperledger-fabric.readthedocs.io/en/latest/security_model.html

2

Hyperledger Fabric — tamper-resistant ledger, world state and historyhyperledger-fabric.readthedocs.io/en/release-2.5/ledger/ledger.html

3

NIST — AES-GCM authenticated encryption and HMAC standardscsrc.nist.gov/pubs/sp/800/38/d/final • csrc.nist.gov/pubs/fips/198-1/final

4

India privacy context — DPDP Act 2023; UIDAI Aadhaar Data Vault guidancemeity.gov.in • uidai.gov.in/images/Circular-No.14_of-2025.pdf